

Topic: Fourier Series

Fourier Analysis is the study and representation of periodic functions

Fourier series are used to represent and approximate periodic functions. This is similar to Taylor Series, except that Fourier Series are not restricted to a neighborhood of a specified point.

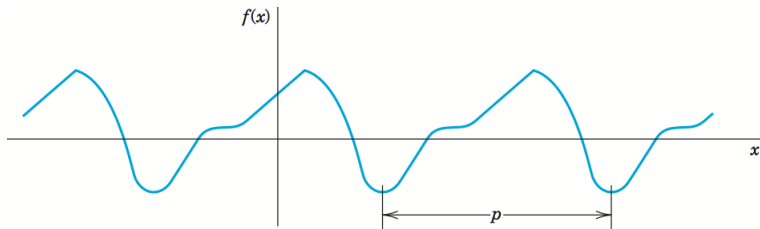
Many physical phenomena are periodic and are usually described in the following two contexts:

- ▷ periodic in space (period)
- ▷ periodic in time (frequency)

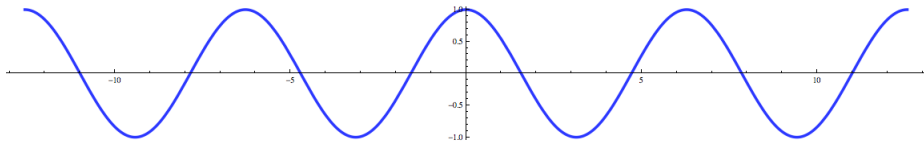
Examples of phenomena that are periodic:

- ▷ Heat on a ring or bar (bar with special boundary conditions)
- ▷ Harmonic Motion: pendulum
- ▷ Wave Motion - periodic in both space (wavelength) and time (frequency)
- ▷ Consumer demand - the demand of many products can be modeled by periodic functions

Defn: A function $f(x)$ is **periodic** with **period** p if



Ex: $f(x) = \cos x$ is periodic with period



Defn: Suppose that $f(x)$ is periodic with period 2π . The **Fourier series** of f is

$$f(x) =$$

where d_n are the complex **Fourier coefficients** of $f(x)$ given by

$$d_n =$$

Using Euler's formula () the fourier series can be written as:

$$f(x) =$$

where a_0, a_n , and b_n are the real **Fourier coefficients** of $f(x)$ given by

$$a_0 =$$

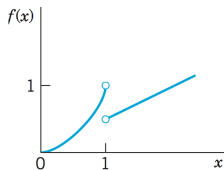
$$a_n =$$

$$b_n =$$

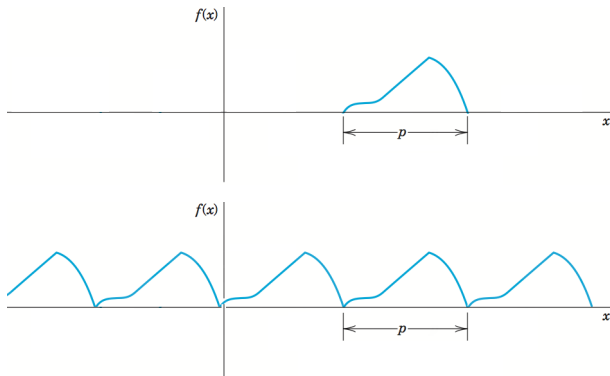
The main goal for this section is to determine either d_n or a_n & b_n .

Notes:

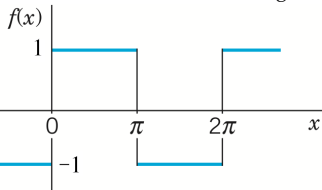
- (1) Fourier series are used to approximate periodic piece-wise continuous functions?



- (2) If a non-periodic function is defined on an interval, we can force it to be periodic by artificially making copies and extending its domain as shown below.



Ex: Find the Fourier Series of the function given by



$$f(x) =$$

$$d_n$$

$$d_0$$

$$(1)$$

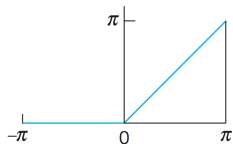
$$(2)$$

$$d_n$$

Therefore,

$$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{inx} =$$

Ex: Find the Fourier Series of the function given by



$$f(x) =$$

